

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 1_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 38.5

Section 1 : Coding

1. Problem Statement

Mandy is a software engineer working on a program to analyze two integers based on specific conditions using a logical operator. She needs to determine if both integers are odd or if at least one of them is divisible by 7.

Depending on the result, she wants to print different messages.

If the condition is met, the program should identify and print the first number that is divisible by 7 or indicate that both numbers are odd. If the condition is not met, the program should print a message indicating the condition was not met, along with the input numbers.

Input Format

The first line of input consists of an integer representing the first input number.

The second line consists of an integer representing the second input number.

Output Format

The output displays "Condition met: " followed by an integer representing the first number divisible by 7, or prints "Both numbers are odd" if the two inputs are odd.

If the condition is not met, it displays "Conditions not met: " followed by the two input integers, separated by a space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 7

14

Output: Condition met: 7

Answer

```
// You are using Java
import java.util.Scanner;
class NumberConditionChecker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int num1 = scanner.nextInt();
        int num2 = scanner.nextInt();

        boolean bothOdd = (num1 % 2 != 0) && (num2 % 2 != 0);
        boolean num1Div7 = (num1 % 7 == 0);
        boolean num2Div7 = (num2 % 7 == 0);

        if (bothOdd) {
            System.out.println("Condition met: Both numbers are odd");
        } else if (num1Div7 || num2Div7) {
            int firstDivisible = num1Div7 ? num1 : num2;
            System.out.println("Condition met: " + firstDivisible);
        } else {
            System.out.println("Conditions not met: " + num1 + " " + num2);
        }
    }
}
```

Status : Partially correct

Marks : 8.5/10

2. Problem Statement:

"Write a program that helps identify the type of a triangle based on the lengths of its three sides. The program prompts the user to input the lengths of sides 'a', 'b', and 'c', and then it classifies the triangle as 'Equilateral' if all sides are equal, 'Isosceles' if two sides are equal, or 'Scalene' if all sides are different. Can you provide the Java code for this task?"

Input Format

The first line of the input is an integer 'a' representing the length of side 'a.'

The second line of the input is an integer 'b' representing the length of side 'b.'

The third line of the input is an integer 'c' representing the length of side 'c.'

Output Format

The program outputs a single line that specifies the type of the triangle: "Equilateral," "Isosceles," or "Scalene."

Sample Test Case

Input: 3

4

5

Output: The triangle is Scalene

Answer

```
// You are using Java
import java.util.Scanner;
class TriangleType {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int a = scanner.nextInt();
        int b = scanner.nextInt();
```

```
int c = scanner.nextInt();  
    if (a == b && b == c) {  
        System.out.println("The triangle is Equilateral");  
    } else if (a == b || b == c || a == c) {  
        System.out.println("The triangle is Isosceles");  
    } else {  
        System.out.println("The triangle is Scalene");  
    }  
}  
}
```

Status : Correct

Marks : 10/10

3. Problem Statement:

Gilbert is tasked with writing a program that checks whether a given integer is an odd number. An odd number is one that cannot be exactly divided by 2. The program should take an integer as input and determine if it is an odd number or not. The task is to implement the logic to check if the provided integer is odd and return the result.

Input Format

The first line of the input contains an integer, "input".

Output Format

The output should display a boolean value, "result," which should be set to true if the input integer is an odd number and false if it is even.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 0

Output: Is the integer odd? false

Answer

// You are using Java

```
import java.util.Scanner;
class OddChecker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int input = scanner.nextInt();

        boolean result = (input % 2 != 0);
        System.out.println("Is the integer odd? " + result);
    }
}
```

Status : Correct

Marks : 10/10

4. PROBLEM STATEMENT:

Julia a mathematician expert is given two integers to find if the second integer is above the average of the first and second integer. Write a program that achieves this using the ternary operator.

Input Format

The first line of input represents the first integer.

The second line of input represents the second integer.

Output Format

The output should be displayed as "Below Average" or "Above Average"

REFER THE SAMPLE TESTCASES FOR THE FORMAT SPECIFICATIONS.

Sample Test Case

Input: 1

1

Output: Below Average

Answer

// You are using Java

```
import java.util.Scanner;
class AverageCheck {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int first = scanner.nextInt();
        int second = scanner.nextInt();

        double average = (first + second) / 2.0;

        String result = (second > average) ? "Above Average" : "Below Average";
        System.out.println(result);
    }
}
```

Status : Correct

Marks : 10/10